



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6173**

L 61, 475, 986, 1, 1505, 48279

AV:	2025-26	Session:	Date:
Sem:	1	Odd MSE October 2025	08-10-2025
Exam Type:	Written	Slot:	C. Code:
		10:00am - 11:00am	CSA13
		Block: 202	Seat No:
		Course Section: 1	2022600018

Name of the candidate. Jiten Ishwar Granwani

Candidate's UID number:

2	0	2	2	6	0	0	0	1	8
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Examination class room number:

2	0	2
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Invigilator's signature with date:  08/10/2025



(To be filled in by the candidate)

EXAMINATION: BE Btech MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE AJML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8/10/2025

COURSE NAME: Deep learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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(Begin answer for each question on a new page)

Q2b) → In linear undercomplete autoencoder latent dimension space is less than input dimension space.

→ This creates a bottleneck in the autoencoder architecture between encoder and decoder.

→ This bottleneck forces the model to learn important features only and it ignores other features. In this way it is an unsupervised learning. In this way, undercomplete autoencoders act like PCA.

→ In overcomplete autoencoders the hidden latent space is greater than the input dimension. Because of this the autoencoder model learns to copy the input image/data.

→ With regularization we solve this by adding different methods.

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(Begin answer for each question on a new page)

Q3a) for RNN gradient is ~~not~~ updated like this

$$\frac{\partial L}{\partial k} =$$

$$\frac{\partial L}{\partial h_{t+1} k} = \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial k} \quad (\text{as with hxi update})$$

$$\text{in RNN, } h_t = \tanh(a_t h_{t-1} + b_t)$$

$$h_t = \tanh(a_t h_{t-1} + b_t) \quad \frac{\partial h_t}{\partial h_{t-1}} = 1 - \tanh^2(a_t h_{t-1} + b_t)$$

eqn. (1)

$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_{t-1}}{\partial h_{t-2}} \times \frac{\partial h_{t-2}}{\partial h_{t-3}} \dots \frac{\partial h_{t-n}}{\partial h_n}$$

In case of LSTM in forget gate if the gate values becomes close to 1 the gradient flows directly to the candidate state and hidden state and thus because of this it doesn't get involved in recursion.

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Q3b) GRU has 2 gates, update, reset

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z)$$

Gr $z_t =$

update gate $z_t = \sigma(W_z[h_{t-1}, x_t] + b_z)$

$$\tilde{c}_t = \tanh(W_c[\sigma(h_{t-1}, x_t) + b_c])$$

~~Let~~

$$h_t = (1 - z_t)h_{t-1} + z_t \tilde{c}_t$$

~~Let~~

~~Let GRU~~ ^{GRU} GRUs have 2 gates update & reset ^{and hidden} same state while LSTM has to deal with 3 gates input, output and forget. Because of this there are more parameters in weight matrices and bias.

- Because of this GRU is ~~more~~ able to capture perform better in simpler task where patterns are less complex

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~~Q1a)~~

Q1a)

~~$$f(w) = \frac{1}{2} \|xw\|$$~~

$$f(w) = \frac{1}{2} \|xw - y\|^2$$

~~$$\frac{\partial f}{\partial y} = \frac{1}{2} \|xw - y\| (0 - 1)$$~~

$$= -\|xw - y\|$$

~~$$\frac{\partial f}{\partial w}$$~~

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Q2) In case of standard 2D convolution the input image would have to be taken again & again to compute the things again & again. In case of depthwise separable convolution we won't need that and it is important as we create feature maps from the same

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